

# Arthur Villemann

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Echandens, Vaud – 1026, Switzerland

## PROFILE

Mechanical engineering Master's student specialising in composite structures, with four years designing, analysing and manufacturing carbon fibre chassis components in Formula Student. Seeking the Composite Design, Simulation and Structure Industrial Placements in motorsport, to work on chassis and bodywork design from laminate definition, analysis through to manufacture.

## EDUCATION

**EPFL** 2026-09 – Present  
*MSc in Mechanical Engineering* Lausanne, Switzerland

**EPFL** 2022-09 – 2026-07  
*BSc in Mechanical Engineering* Lausanne, Switzerland

◦ 1<sup>st</sup> year GPA: 4.29/6.00    2<sup>nd</sup> year GPA: 4.36/6.00    3<sup>rd</sup> year GPA: 4.64/6.00

## EXPERIENCE

**EPFL Racing Team** 2023-09 – Present  
*Chassis Engineer (2023-09 – 2025-09)* Lausanne, Switzerland

- Designed critical aluminium inserts in CATIA; ran FEA in 3DEXPERIENCE for pull-out and compressive strength in carbon fibre laminate, correlating predictions to within 2% of coupon test results.
- Reduced chassis mass by 2.5 kg (-8%, from 31 kg) through insert optimisation.
- Manufactured the composite chassis and aerodynamic kit for the team's fifth and sixth cars.

*Chassis Research Engineer (2025-09 – 2026-09)*

- Development of the team's first carbon fibre rims.

*Chassis Team Leader (2026-09 – Present)*

- Leading the Chassis division: 10 engineers across 6 subsystems, CHF 40,000 budget.

**EPFL / Students 4 Students** 2023 – 2025  
*Teaching Assistant & Lecturer* Lausanne, Switzerland

- TA for *Programmation pour ingénieur* (Prof. C. Salzmann) and for the integration week (Analysis, Linear Algebra, Physics, Electronics); lecturer for Analysis 1 (2025).

## PROJECTS

**Design and Simulation of a Carbon Fibre Rim for a Formula Student Vehicle** 2026-02 – 2026-06

- Led development of the team's first carbon fibre rim, replacing the 1.8 kg aluminium unit with a 0.7 kg composite design (-61% per wheel, 4.4 kg unsprung mass across the car).
- Sized the laminate in FEM to a 0.2°/g camber stiffness target (FEM result 0.267°/g); rig validation scheduled for April 2027.

**Thermal and Structural FEM Analysis of an Aerospike Combustion Chamber** 2026-02 – 2026-06

- Four FEM studies in 3DEXPERIENCE comparing two regeneratively cooled geometries of a Inconel 718 aerospoke at high chamber pressures.
- Second iteration : film cooling, resized channels, reprofiled spike : cut peak wall temperature from 2389 K to 1520 K (-36%), von Mises stress from 178 to 138 MPa (-22%) and maximum displacement from 75 to 30 µm (-60%).

## SKILLS

- **Composites:** Laminate design & layup, aluminium insert design, bonded joints, sandwich panels, vacuum bagging, tooling design, coupon testing, experimental test planning
- **Simulation:** FEA/FEM (3DEXPERIENCE), thermal and structural analysis, composite failure analysis, mesh convergence, dynamic modelling
- **CAD & Manufacturing:** CATIA V5, 3DEXPERIENCE, Fusion 360, technical drawing, 3D printing, lathe & mill machining, Git
- **Programming:** Python, MATLAB, C, LabVIEW
- **Languages:** French (native), English (fluent), German (intermediate)